Project Documentation: Process Maps and Literature Validation

for the Rosenthal Melt-Pool Calculator

Adeleke Olorunnisola Oyeyemi

Department of Mechanical Engineering, Federal University of Technology, Akure, Nigeria

Source code: https://github.com/Olorunnisola01/rosenthal-meltpool

Interactive demo: https://rosenthal-meltpool.streamlit.app

***Abstract*—This document describes Project 2 of a six-month research-preparation programme in metal additive manufacturing (AM) process modelling, building directly on Project 1 (a Python/Streamlit calculator implementing the Rosenthal moving point-heat-source solution, documented separately). Project 2 has two parts: a power-velocity parameter sweep that generates process maps for four L-PBF alloys, and a validation of the calculator's predictions against a real published single-track dataset. The validation is the more consequential result: at the source study's own assumed absorptivity, the model underpredicts every published case, and a calibration search shows this cannot be corrected by absorptivity alone for most cases. Combined with a process-map finding — that the predicted depth-to-width ratio is exactly 0.5 everywhere, for every material, confirmed here across the full swept parameter range rather than at isolated points — this document establishes the quantitative boundaries within which the Project 1 calculator's output should be trusted, and motivates the heat-source modelling choices for later, more physically complete work.**

I. Introduction

Project 1 established that the Rosenthal moving point-heat-source solution predicts a melt pool whose depth is always exactly half its width, for any material or process parameters — a mathematical identity of the model, confirmed at a handful of worked examples. Two questions follow directly from that finding, and both required going beyond single-point evaluation: does the identity hold everywhere across the process window, or only at the specific parameter combinations tested; and, more importantly, how do the model's absolute predictions compare against a real measured single-track dataset, not just against each other. This project answers both questions by (i) sweeping the calculator over a full power×velocity grid for all four preset alloys, and (ii) validating predictions against a published dataset with independently measured melt-pool dimensions.

II. Parameter Sweep Methodology

The core package's `sweep.py` module evaluates `melt_pool_dimensions` at every point on a power×velocity grid, holding absorptivity and preheat temperature fixed, and returns width, depth, length, and aspect ratio (depth/width) as 2D arrays. Points where the transcendental root-finder fails to converge (sub-threshold parameters that produce no melt pool at all) are recorded as `NaN` rather than raising an error, so a sweep can span parameter combinations where some points are physically sub-threshold without halting the whole grid.

For the process maps in this document, power was swept from 50 to 400 W and velocity from 200 to 2000 mm/s, each in 45 steps, at a fixed absorptivity of 0.4 and preheat of 300 K, matching the Project 1 calculator's default operating point.

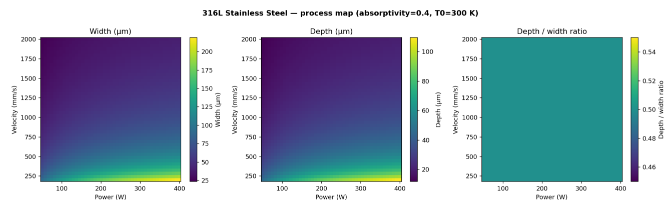

Fig. 1: 316L process map: width, depth, and depth/width ratio across the full power-velocity grid. The third panel is visually uniform because it is uniform — every point evaluates to 0.5, to within floating-point precision.

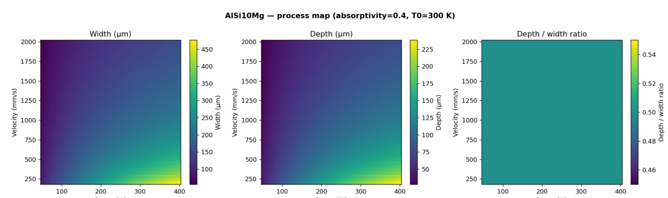

Fig. 2: AlSi10Mg process map at identical sweep ranges. Peak width reaches roughly 470 µm, over twice 316L's roughly 220 µm peak in Fig. 1, at identical power and velocity — a direct visual confirmation that thermal conductivity, not process parameters, sets the overall scale of the melt pool.

III. Process Maps

Figs. 1 and 2 show the swept process maps for 316L and AlSi10Mg. Both alloys show the expected qualitative trend — width and depth increase with power and decrease with velocity — but the two materials occupy entirely different absolute scales, consistent with AlSi10Mg's roughly 5× higher thermal conductivity. The third panel in each figure, depth/width ratio, is the more important result: it is flat — not approximately flat, but numerically 0.5 at every grid point tested, for both materials and across the full power-velocity range. This confirms, by direct computation rather

TABLE I: Predicted vs. measured melt-pool dimensions, Zhang et al. (2024)

Case	P (W)	v (mm/s)	W_{pred} (µm)	W_{meas} (µm)	D_{pred} (µm)	D_{meas} (µm)
N01	260	520	105.8	114.0	52.9	180.0
N04	260	1470	48.7	94.0	24.4	61.0
N05	260	2200	35.6	83.0	17.8	41.0
N06	440	1470	54.7	98.0	27.4	104.0

TABLE II: Calibrated absorptivity search (bounds: 1%–500%)

Case	Calibrated for width	Calibrated for depth
N01	0.658	no solution
N04	no solution	no solution
N05	no solution	no solution
N06	no solution	no solution

than by the algebraic argument alone, that the semicircular cross-section identified in Project 1 is not an artifact of the specific parameter values used there; it holds everywhere this model can be evaluated. A dedicated regression test locks this property into the package's test suite across all four alloys, so any future change to the model that broke it would be caught immediately.

IV. Literature Validation

A. Dataset and Method

Model predictions were compared against four published bare-plate single-track measurements for 316L stainless steel, from Zhang *et al.* (2024) [1], Table 3. The source study reports laser power, scan velocity, measured melt-pool width, and measured melt-pool depth for each case, and states an assumed absorptivity of 0.5 and a 100 µm laser spot diameter in its own numerical model. Predictions were generated with the same absorptivity (0.5) and a preheat of 300 K (room temperature, the conventional default for bare-plate experiments; not stated explicitly in the source paper).

B. Results

Table I gives predicted and measured width and depth for all four cases. The model underpredicts every single case, in both dimensions, ranging from a -7.2% width error (N01) to a -73.7% depth error (N06). Measured aspect ratios (depth/width) for these four cases are 1.58, 0.65, 0.49, and 1.06 respectively — directly demonstrating the range of real melt-pool shapes the model's fixed 0.5 ratio (Section III) cannot represent.

C. Absorptivity Calibration Search

Because absorptivity is the one input not fixed by material properties, a calibration search was run for each case: holding power, velocity, and material fixed, find the absorptivity that makes predicted width (or depth) exactly match the measured value, searching up to 500% — a deliberately unphysical upper bound, used to distinguish "the guessed absorptivity was wrong" from "no absorptivity closes this gap."

Table II shows that depth cannot be matched for *any* of the four cases at any absorptivity up to 500%, and width can only be matched for one case (N01, at a plausible 65.8%). For the other three cases, the gap between prediction and measurement persists even at an absorptivity five times larger than physically possible.

D. Interpretation

The predicted pool sizes in these cases (35–115 µm) are the same order of magnitude as the 100 µm laser spot diameter the source study reports. The point-source assumption underlying the Rosenthal solution requires the resulting pool to be much larger than the source itself; these cases sit at or below that threshold, so the assumption is not satisfied, independent of what absorptivity value is chosen. This is a more precise, falsifiable statement than the qualitative caveat given in Project 1 — it is now backed by a specific numeric search showing where and how badly the assumption breaks down for a real dataset.

N01 is the case closest to being matched (width matchable at a plausible absorptivity), and it is also the case with the highest measured aspect ratio (1.58, i.e. keyhole-influenced) and the largest absolute pool size of the four — consistent with the point-source assumption holding somewhat better as pool size grows relative to spot size, even though depth remains unmatched because keyhole-mode depth enhancement is outside what any conduction-only model can represent.

V. Implications for Paper 2

These results directly shape the analytical melt-pool modelling manuscript planned as Paper 2. The process maps (Section III) supply the standard process-window figures expected in that paper's methodology section. The validation and calibration search (Section IV) supply its most citable content: a quantified, falsifiable account of where and why the classical point-source model fails on real data, rather than a restatement of textbook caveats. The natural next methodological step — not undertaken here — is comparison against a heat-source model that removes the point-source assumption, such as a finite Gaussian surface flux or Goldak's double-ellipsoidal source [2], to test whether either closes the gap identified in Table I without sacrificing the closed-form speed that makes the Rosenthal model useful for process-window screening in the first place.

VI. Conclusion

Project 2 extends Project 1's single-point calculator into a swept process-mapping tool and, more importantly, into a quantitatively validated one. Two findings carry forward into the accompanying paper: the depth/width$= 0.5$ identity holds across the entire explored process window for every alloy, not just at isolated points; and a calibration search against real published data shows the model's absorptivity is not always sufficient to explain observed discrepancies, with the root cause traced to a specific, checkable condition (pool size relative to laser spot size) rather than left as an unquantified caveat.

References

[1] S. Zhang *et al.*, “Understanding melt pool behavior of 316L stainless steel in laser powder bed fusion additive manufacturing,” *Micromachines*, vol. 15, no. 2, p. 170, 2024, doi: 10.3390/mi15020170.

[2] J. Goldak, A. Chakravarti, and M. Bibby, “A new finite element model for welding heat sources,” *Metallurgical Transactions B*, vol. 15, no. 2, pp. 299–305, 1984.